



vehicle dynamics solutions

**OptimumG Vehicle Dynamics Seminar Applied to Vehicle Concept, Simulation,
Testing and Development**

4-Day Seminar

Seminar Content Overview

Seminar Objectives

It's about connecting the dots. Through a vehicle dynamics course, applied to vehicle concept, design, simulation, testing and development specifically tailored for automotive engineers, OptimumG can help participants come away with a better understanding and appreciation of how all parts of the vehicle must work in cohesion to have an outstanding result.

Previous seminar attendees have reported an increase in the quality of communication among the individual team members as well as new perspectives on already applied principles in vehicle design and development. Provided within this document is a background of OptimumG along with an overview of a proposed course of action for a vehicle dynamics seminar for the engineers.

OptimumG recommends a four-day program specifically focusing on race car design and development effects on car grip, balance, control, stability and overall racetrack performance and consistency.

This four-day program is a blend of academic course and a practical workshop inviting questions and answer as well as interactions between the participants.

It will be taught by OptimumG president Claude Rouelle who will bring his race car engineering expertise and practical on-track examples simulation, data acquisition and data bases management.

Company Background

A vehicle dynamic consulting firm, OptimumG helps clients enhance their understanding of vehicle concept, simulation, design, testing, development, and analysis of on-board acquired data from truck and of-road to race car vehicle through seminars, consulting, and software development.

OptimumG founder and president, Claude Rouelle, has conducted over 380 seminars for more than 14,000 engineers in 34 different countries and has been teaching vehicle dynamics and vehicle design in several universities.

OptimumG has over 280 years of combined automotive engineering experience ranging from commercial vehicles to high performance and race cars.

Our engineers have provided in-house seminars, training programs, simulation products and services for companies such as Dunlop, Michelin, Goodyear, Bridgestone-Firestone, Ferrari, Peugeot, Toyota, General Motors, Chrysler, ZF Sachs...just to name a few.

At OptimumG, we have the engineering expertise, knowledge, and perspectives to provide automotive engineers with the continuous performance their organization is striving to achieve.

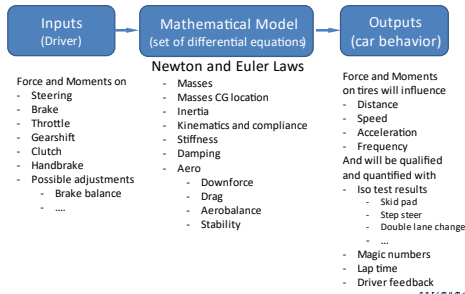
Seminar Content

(Description illustrated with some of the seminar slides)

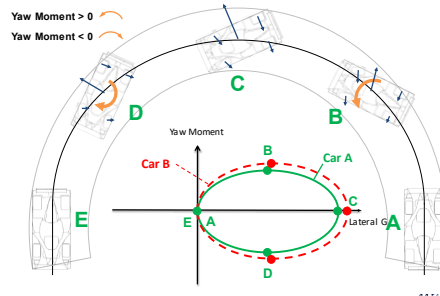
Day I

Performance definition. What makes a car go fast in a corner, and what allows a car to get in and out of a corner quickly? As a short introduction to the seminar, the participants and Claude Rouelle will share their perspectives on steady state and transient performance definition.

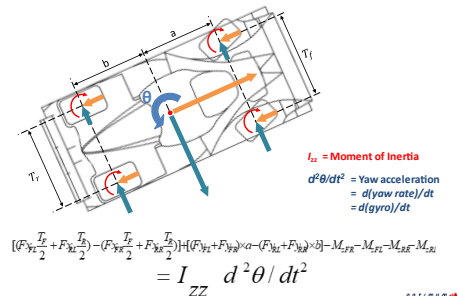
How to Study Vehicle Dynamics



Yaw Moment vs. Lateral Acceleration

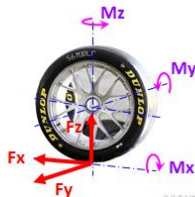


Transient Basics: Yaw Moment Equation

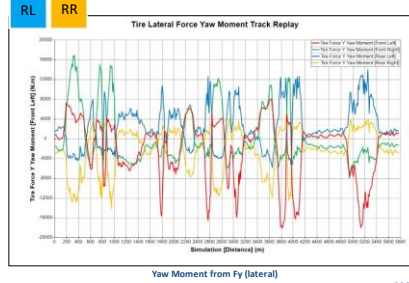


The 12 Causes of Yaw Moment

Wheel force transducers measure all tire forces at the wheel center
The yaw moment contribution from each tire can be calculated from these forces



The 12 Causes of Yaw Moment



Yaw Moment Needed

Go from the outer skid pad (1) to the inner skid pad (2) (or the other way around) at maximum lateral acceleration without over or understeer.

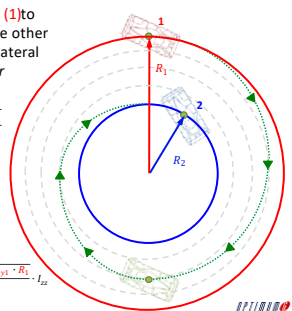
$$\Delta r = \frac{R_1 \cdot \sqrt{A_{y2} \cdot R_2 - R_2 \cdot \sqrt{A_{y1} \cdot R_1}}}{R_2 \cdot R_1}$$

$$\Delta t = t_2 - t_1 \text{ and } \dot{r} = \frac{dr}{dt} = \frac{\Delta r}{\Delta t}$$

$$\dot{r} = \frac{R_1 \cdot \sqrt{A_{y2} \cdot R_2 - R_2 \cdot \sqrt{A_{y1} \cdot R_1}}}{R_2 \cdot R_1 \cdot (t_2 - t_1)}$$

Yaw Moment = $\dot{r} \cdot I_{zz}$

$$\text{Yaw Moment} = \frac{R_1 \cdot \sqrt{A_{y2} \cdot R_2 - R_2 \cdot \sqrt{A_{y1} \cdot R_1}}}{R_2 \cdot R_1 \cdot (t_2 - t_1)} \cdot I_{zz}$$

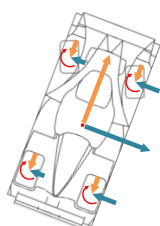


Tires Understand “why” and “how much” the grip, balance and performance of a car is decided by the forces and moments generated in the tire contact patch. We will study the influence of lateral and longitudinal slip, vertical load, camber, speed, pressure and temperature as well as these combined inputs on lateral, longitudinal and combined accelerations, balance and driver feedback. We will also cover in-lab and on-track tire testing, tire modeling as well as the benefits and limitations of tire model as input of simulation software:

The Importance of the Tire

Most handling issues are primarily about tire forces and moments

- Yaw moment
- Agility / Stability / Crash Avoidance
- Performances
 - Cornering
 - Braking
 - Combined accelerations
- Indispensable in car and tire design
- Car and tire simulation
- Car and tire development
- ESP / ABS / Traction control

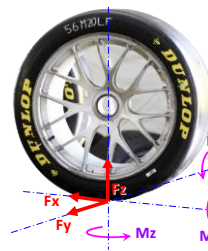


Tire Measurements

Tire Testing Overview

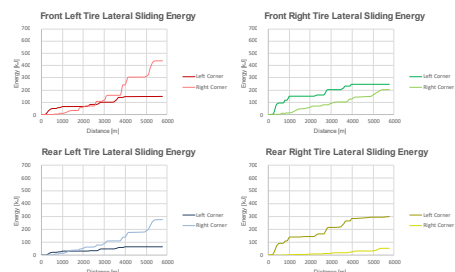


Tire Forces and Moments

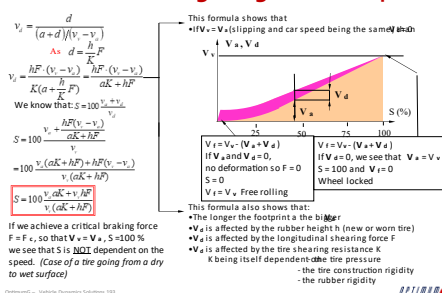


- 2 Forces Fy and Fx and 3 Moments Mx, My and Mz are a function of:
- Slip angle
 - Slip Ratio
 - Vertical Load
 - Inclination Angle
 - Speed
 - Pressure
 - Temperature
 - Rim Design
 - Tire Wear
 -

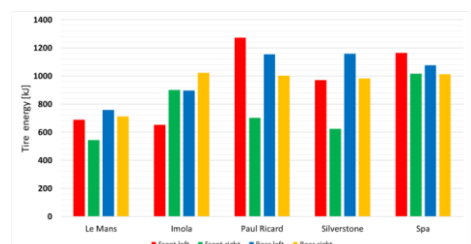
Lateral Sliding Energy - Sebring



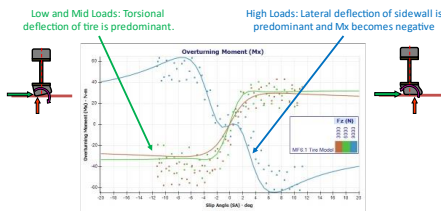
Understanding Longitudinal Slip



Comparing Energies



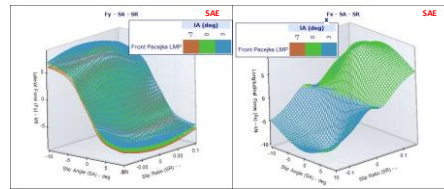
Overturning Moment Caused by Fy



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Inclination Angle Effects on Fy and Fx



Lateral:
The surfaces are shifted horizontally.

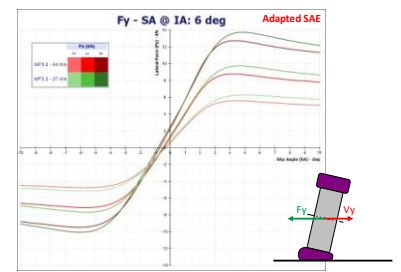
Longitudinal:
More camber gives us less longitudinal force over the entire range.

Note: Must use Pacejka MF5.2 or newer to see these effects.

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e) Speed effect on Fy



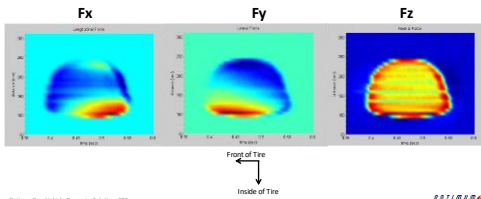
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Force Matrix Sensor

This type of sensor can help increase understanding on how the tire operates during driving, braking and cornering situations.

Measuring an actual tire has the benefit that it can capture complex frictional behaviors, which currently is difficult to model in FEA.

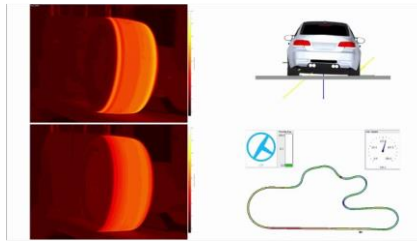


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Track Replay Testing

Consists of reproducing the operating conditions of the tire at the track in a Flat-Track machine



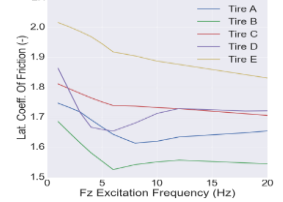
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Avg. Lat. CoF. vs. Excitation Frequency

Fz Stat. = 4000 N; Fz Amp. = +/- 650 N; SA = 5°; V = 40 km/h; IA = 0°; P = 1.90 bar; 2.1

- Avg. Lat. CoF. is the Avg. Fy / Avg. Fz during oscillation
- A curve is fit for each tire using a 30% data range LOWESS approach



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Steady State Fundamentals The review of simple equations of motion will focus on first definition and quantification of understeer and oversteer. Practical meaning of these equations will help us to evaluate the contribution of CG slip angle, yaw rate, and steering to lateral acceleration and yaw moment.

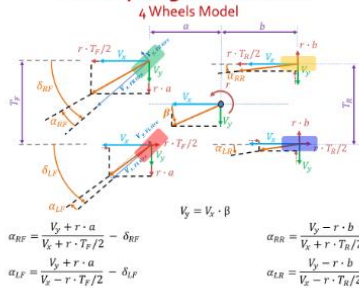
The 3 causes of slip angle: δ , β and r



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Tire slip angle calculation

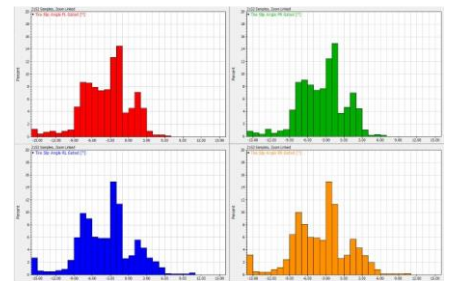


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Tire Slip Angle Histogram

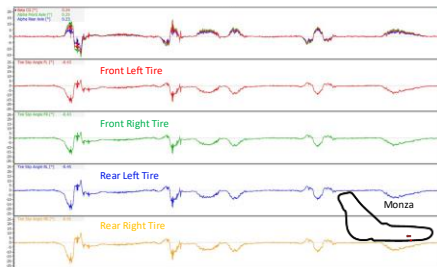
Slip angles gated by absolute lateral G greater than 0.5



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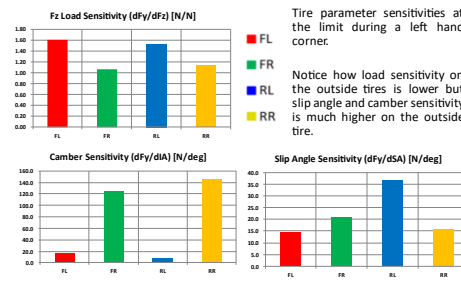
Example of Slip Angle Sensor Data



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Tire Parameters Sensitivities



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Understeer and Oversteer

- The yaw acceleration is a function of your car yaw inertia and your car yaw moment
- The yaw inertia needs to be measured on a inertia bench
- The yaw moment is a function of your tire longitudinal and lateral forces, your wheelbase your front and rear tracks.
- The ideal yaw acceleration is 0 on a skip pad (steady state)
- The ideal yaw acceleration you need in transient depends on the speed the radius of the corner you are going through.
- As the speed and the radius never stop changing, the ideal yaw acceleration (and therefore the ideal yaw moment) never stop changing either

$$\ddot{\Theta} = + \frac{d\dot{\Theta}}{dt} \quad \ddot{\beta} = \frac{d}{dt} \frac{\text{Lateral G}}{V_x} + \ddot{\beta}$$

Yaw Moment = Inertia_{yaw} x $\ddot{\Theta}$

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Aerodynamics After a review of aerodynamics basics, we will focus on the understanding of aero-maps static and dynamic ride height settings, and how to integrate them into the design of a suspension and the choice of suspension stiffness and damping.

Aerodynamics

"Aerodynamics are for people who can't build engines"

Enzo Ferrari

Automotive Aerodynamics

Aero Mapping: Single Seater Example 1

Aerobalance (% front downforce / total downforce)

		Rear Ride Height (mm)					
		3.0	13.0	23.0	33.0	43.0	53.0
Front Ride Height (mm)	33.0	34.07	32.98	33.01	33.48	35.14	36.54
	23.0	36.05	35.28	35.27	36.20	37.53	39.08
	13.0	38.80	37.87	37.83	38.33	39.58	41.21
	3.0	41.26	40.66	40.92	41.67	43.37	45.30

Rule of thumb

For one step of front ride height variations we need 2 or 5 steps of rear ride height variations to keep the same aerodynamic balance.

In the specific case of the LMP car used in this seminar, we have only a 1 to 1.5 rear / front ride height variation ratio. However, most races follow the Rule of Thumb.

You thought aeromaps were easy?

Transient Aeromaps Height Variation

Kinematics See why poorly designed kinematics cannot be “patched” by springs, antiroll bars and dampers. From the design to on-track testing and racing, understanding the effects of kinematics is essential to the efficient use of race tires. Examples of tire model exploitation for suspension camber variation and steering kinematics design will be given. With examples of K&C measurements and tire models we will also explain the essential differences between kinematic and force roll and pitch centers.

Inclination Angle Effects on Fy and Fx

Camber Static Setup or Camber Variation ?

Before OptimumG

After OptimumG

Steering Induced Diagonal Weight Transfer

We will look at the changes in ride height and vertical load (Fz) as a function of steering input. This can have a significant effect on the Yaw Moment of the vehicle. Therefore, if these effects can be exploited they can be an effective tuning tool

Vertical dynamics, spring, ARBs, and damper determination

Purpose of Springs, ARBs and Dampers

	Ampl. (mm)	Freq. (Hz)	Speed (mm/s)
SM k_{spring} @ wheel Damper @ wheel	5 - 50	1.5 - 7.0	< 25 - 50
NSM k_{tire} carcass	2 - 10	10 - 40	< 45 - 85
Tire Carcass k_{tire} compound	1 - 2	100 - 200	> 100 - 150

Simulation : Frequency analysis

The mean square (MS) value is representative of the transmissibility over all the frequencies.

A given suspension setup is a compromise between suspension movement and tire load variation.

Time vs suspension transmissibility

Increased Damping

$c = 2000 \text{ Ns/m}$

$c = 200 \text{ Ns/m}$

$c = 800 \text{ Ns/m}$

0.5 Hz to 30 Hz frequency range for RMS optimization

$m_1 = 50 \text{ kg}$
 $m_2 = 8 \text{ kg}$
 $k_1 = 200 \text{ N/mm}$
 $c_1 = 200 \text{ Ns/m}$
 $k_2 = 10 \text{ N/mm}$
 $k_3 = 20 \text{ N/mm}$
 $k_4 = 30 \text{ N/mm}$
 $c = 200, 300, 400, 500, 600, 800, 1200, 2000 \text{ Ns/m}$

Car Analysis

Roll Gradient*

(*) Roll Gradient: Stiffness of the car in roll [deg/g]

Wheel Travel FL [mm] = "Damper Pos FL Filter" [mm] * "Motion Ratio Front Damper" [-]

Roll Angle Front [deg] = atan("Wheel Travel FL" [mm] / "Wheel Travel FR" [mm]) / "Track Front" [mm]

Lateral Weight Transfer Basics



```

graph TD
    A[Weight Transfer] --> B[Non-Suspended Weight Transfer]
    A --> C[Suspended Weight Transfer]
    C --> D["Geometric" Weight Transfer]
    C --> E["Elastic" Weight Transfer]
            
```

Anti-Roll Stiffness Summary

	Anti-Roll Torque	Percentage for Axle	Percentage of total
Front Springs	4236.60 Nm/deg	38.98 %	25.18 %
Front ARB	6631.19 Nm/deg	61.02 %	39.98 %
Front (Springs + ARB)	10867.80 Nm/deg		64.59 %
Rear Springs	3213.10 Nm/deg	53.93 %	19.10 %
Rear ARB	2745.29 Nm/deg	46.07 %	16.32 %
Rear (Springs + ARB)	5958.39 Nm/deg		35.41 %

Anti-Roll Stiffness Distribution = $\frac{\text{Front AntiRoll Stiffness}}{\text{Front AntiRoll Stiffness} + \text{Rear AntiRoll Stiffness}} \times 100$

$$= \frac{10867.80}{10867.80 + 5958.39} \times 100$$

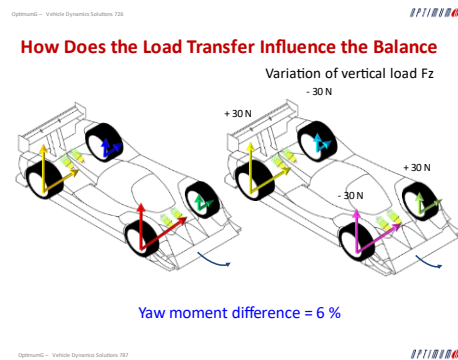
$$= 64.59\% \rightarrow \text{MAGIC NUMBER!}$$

WT Summary

	FL	FR
Static Corner Load	2460.40 N	2460.40 N
NS WT	+ 480.14 N	+ 480.14 N
S "Geometric" WT	- 417.43 N	+ 417.43 N
S "Elastic" WT	- 2604.40 N	+ 2604.40 N
Aero Load	+ 2452.31 N	+ 2452.31 N
Dynamic Corner Load	1410.74 N	8369.45 N
	1455.97 N	

	RL	RR
Static Corner Load	2591.75 N	2591.75 N
NS WT	+ 531.75 N	+ 531.75 N
S "Geometric" WT	- 629.21 N	+ 629.21 N
S "Elastic" WT	- 1446.43 N	+ 1446.43 N
Aero Load	+ 2721.34 N	+ 2721.34 N
Dynamic Corner Load	2705.70 N	7966.29 N
	2659.90 N	

With stiffer ARB at the rear



The reasons for different inside and outside suspended geometric and elastic load transfer

Legend:

- Static weight
- Aero downforce
- Non suspended load transfer
- Suspended geometric load transfer
- Suspended elastic load transfer

Why different?

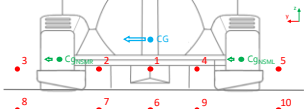
Why different?

Jacking Force Calculation

Roll Center Position Influence

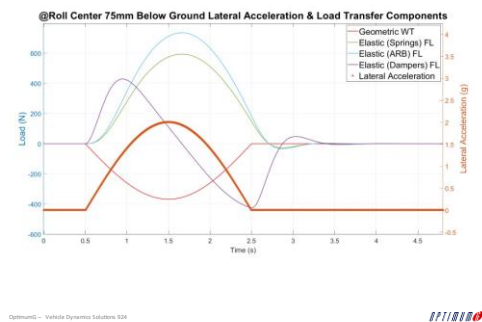
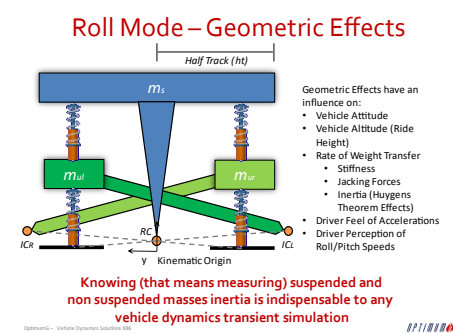
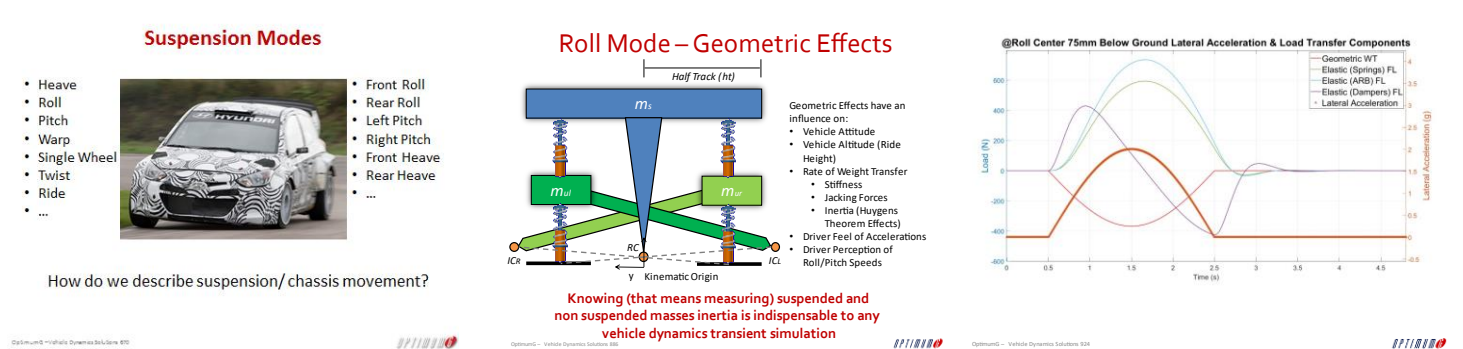
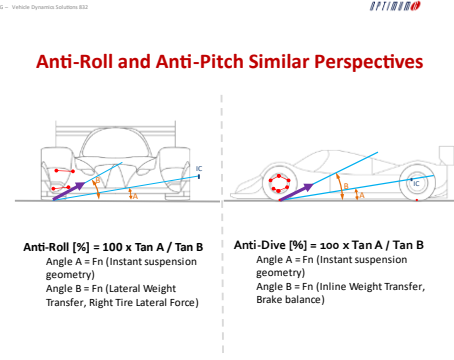
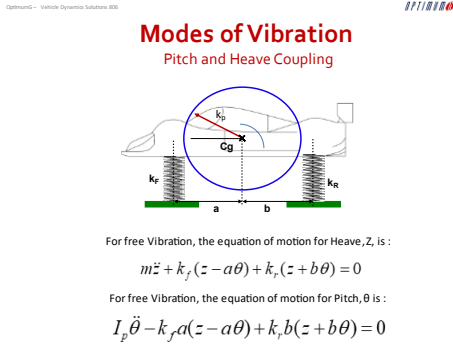
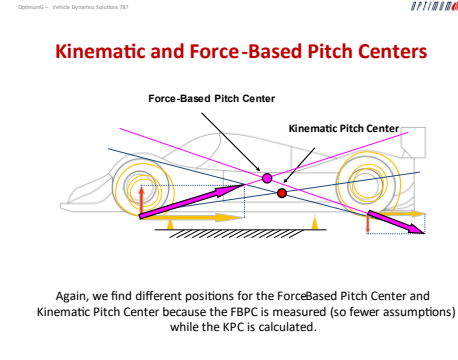


Case	S				S			
Left	Left	Right	Left	Right	Left	Right	Left	Right
Ref. Lift. Pres.	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Ref. Chassis	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
C. Lift. Pres.	-1300.00	-1450.00	-1300.00	-1450.00	-1300.00	-1450.00	-1300.00	-1450.00
C. Chassis	150.79	210.05	150.79	210.05	200.69	278.26	210.79	289.39
Ref. Roll Rate	-273.95	-170.05	-322.00	-839.31	-189.75	-400.00	-200.00	-557.12
Ref. Roll Stiff	547.48	524.52	524.48	524.52	587.72	324.48	524.48	587.72
Ref. Roll Stiff	58.87	13.33	58.87	13.33	58.87	13.33	58.87	13.33
Ref. Roll	439.71	855.79	1099.02	324.59	322.01	299.22	278.25	525.60
Ref. Roll	127.58	200.00	200.00	200.00	200.00	200.00	200.00	200.00
Ref. change from JF	4.33	3.79	-0.00	-0.00	8.68	-2.23	-2.23	-2.23

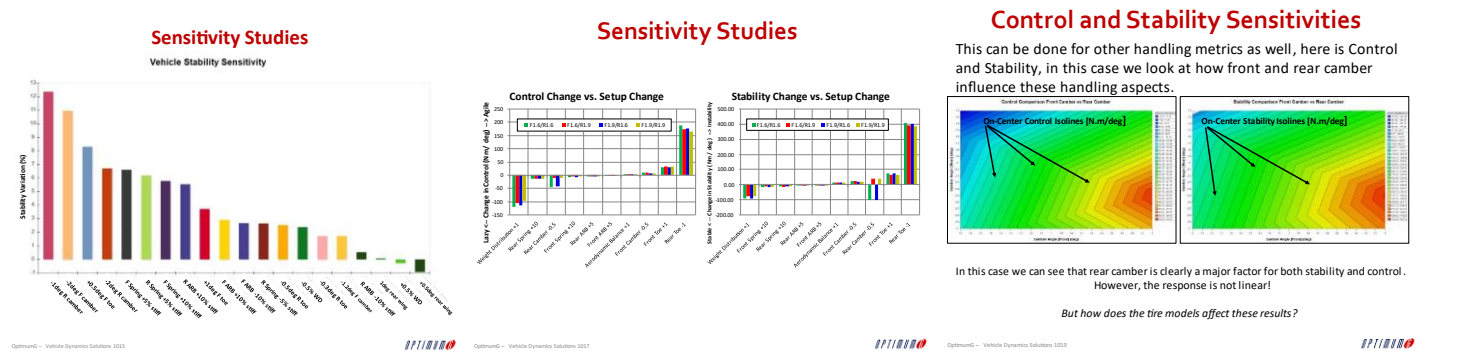
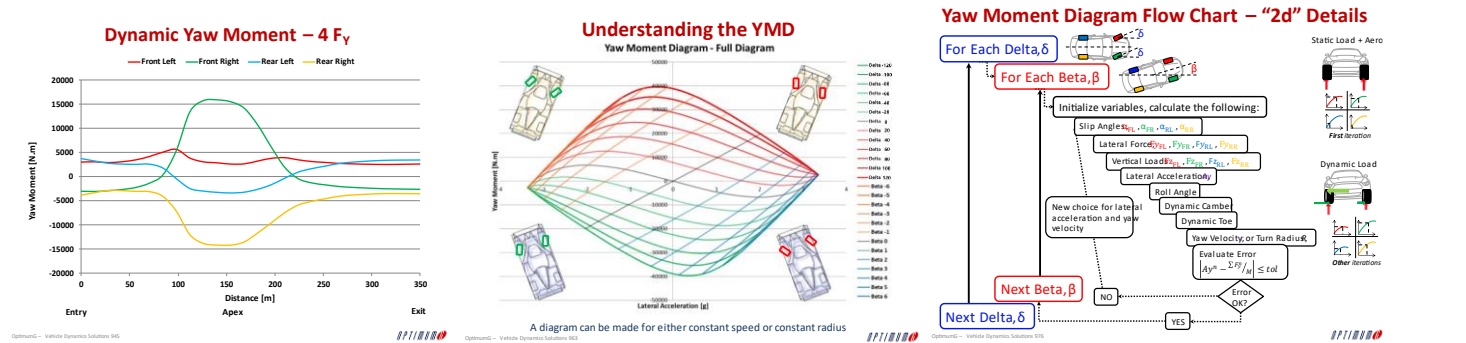








Balance and Grip Compromise: Simulation and on-Track Realities. We will share with you how the yaw moment vs. lateral acceleration method can successfully be applied by integrating 3 different inputs: simulation results, objective analysis from acquired data and subjective driver appreciations through specific driver debriefing method. With scientific demonstration and real examples, we will explain how the choice of the car setup parameters and the driving style both influence the car performance and response for given circuits and given corner segment.

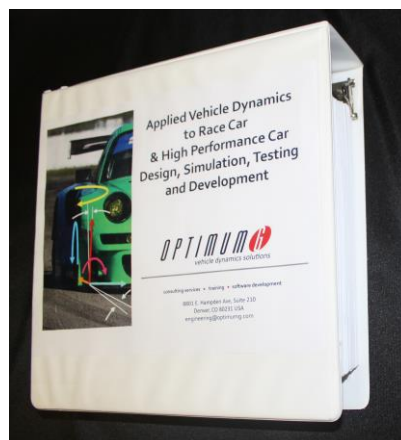


Summary

With a balanced combination of theory refresh and shared experience/perspectives, the Applied Vehicle Dynamics to Race Car Design and Development OptimumG Seminar will help automotive and race engineers to:

- Acquire new knowledge and perspectives,
- Have many of their questions answered,
- Develop new ideas for performance understanding,
- Appreciate the success and the challenges of each other.

A 1000 slides binder will be delivered to each seminar participant.



For information on Schedule and Price visit

<https://optimumg.com/avd-seminar/>
<https://optimumg.com/calendar/>

<https://students.optimumg.com/our-seminars/>
<https://students.optimumg.com/seminar-calendar/>

Miscellaneous

Material: Binder about 1000 pages for each participant.

OptimumG never shares any electronic version of its seminars.

This seminar usually runs daily from 8:00 to at least 20:00 with several morning and afternoon 15-minute short breaks and a one-hour lunch.

The material will be taught in English, by Claude Rouelle, OptimumG president, who has 43 years of experience in vehicle dynamics and over 380 similar seminars and university courses.

The power point slides, and the binder pages are in English.
